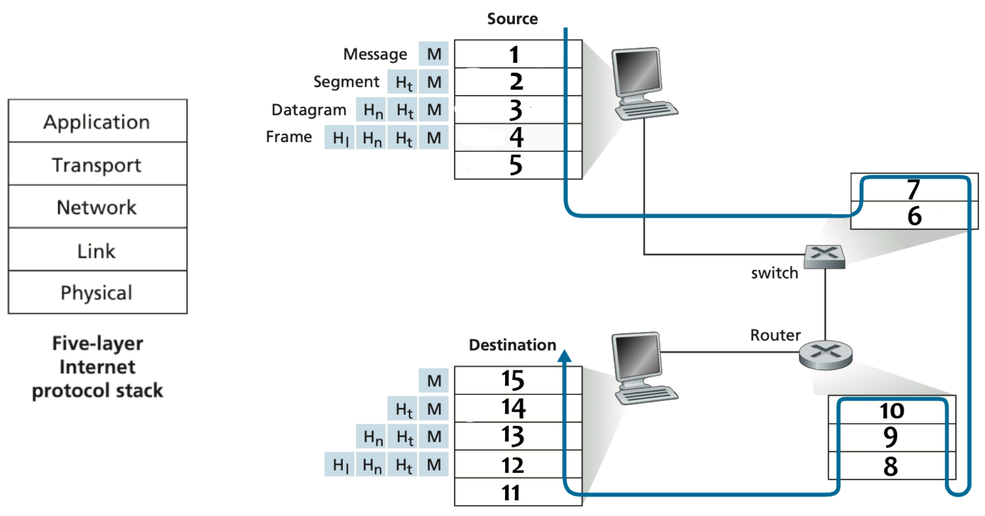
INTERACTIVE END-OF-CHAPTER EXERCISES

THE IP STACK AND PROTOCOL LAYERING

In the scenario below, imagine that you're sending an http request to another machine somewhere on the network.



QUESTION LIST

1. What layer in the IP stack best corresponds to the phrase: 'passes frames from one node to another across some medium'

2. What layer in the IP stack best corresponds to the phrase: 'bits live on the wire'

3. What layer in the IP stack best corresponds to the phrase: 'moves datagrams from the source host to the destination host'

4. What layer in the IP stack best corresponds to the phrase: 'handles messages from a variety of network applications'

5. What layer in the IP stack best corresponds to the phrase: 'handles the delivery of segments from the application layer, may be reliable or unreliable'

6. What layer corresponds to box 1?

7. What layer corresponds to box 2?

8. What layer corresponds to box 3?

9. What layer corresponds to box 4?

10. What layer corresponds to box 5?

11. What layer corresponds to box 6?

12. What layer corresponds to box 7?

13. What layer corresponds to box 8?

14. What layer corresponds to box 9?

15. What layer corresponds to box 10?

16. What layer corresponds to box 11?

17. What layer corresponds to box 12?

18. What layer corresponds to box 13?

19. What layer corresponds to box 14?

20. What layer corresponds to box 15?

SOLUTION

1. The given phrase corresponds to the Link Layer.

2. The given phrase corresponds to the Physical Layer.

3. The given phrase corresponds to the Network Layer.

4. The given phrase corresponds to the Application Layer.

5. The given phrase corresponds to the Transport Layer.

6. Box 1 is the Application Layer.

7. Box 2 is the Transport Layer.

8. Box 3 is the Network Layer.

9. Box 4 is the Link Layer.

10. Box 5 is the Physical Layer.

11. Box 6 is the Physical Layer.

12. Box 7 is the Link Layer.

13. Box 8 is the Physical Layer.

14. Box 9 is the Link Layer.

15. Box 10 is the Network Layer.

16. Box 11 is the Physical Layer.

17. Box 12 is the Link Layer.

18. Box 13 is the Network Layer.

19. Box 14 is the Transport Layer.

20. Box 15 is the Application Layer.

QUESTION 1 OF 20

What layer in the IP stack best corresponds to the phrase: 'passes frames from one node to another across some medium'

Answer

CHECK | HINT | SKIP

We greatly appreciate the work of John Broderick (UMass '21) in helping to develop these interactive problems.

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